

Where the questions come from

Every question is original and written for Brain Blast. They are aligned to Curriculum for Excellence (CfE) Second and early Third level, set in Scottish contexts, and checked automatically on every build. This is the full inventory, by subject and source file.

Two kinds of source

- **Generators** procedural build maths questions from templates with seeded random numbers. The same seed always gives the same question, and every one scales across **Easy / Standard / Hard** tiers.
- **Item banks** hand-written are fixed items with stable ids: maths word problems, spelling words, grammar, vocabulary and reading passages. Review history is tracked against the id, so rewording an item never loses a child's progress.

Curriculum alignment

- **Numeracy & Mathematics:** number, money and measure; shape, position and movement; information handling; early algebra.
- **Literacy & English:** tools for writing (spelling, grammar, punctuation), reading for understanding, and vocabulary.
- Scots is taught as a language alongside Standard English, never labelled as slang.

■ Maths · generators SRC/CURRICULUM · TIERED

TOPIC	CfE ORGANISER	SOURCE	STYLES
Place value & rounding	Number, money & measure	maths.js	multi
Factors, multiples & primes	Number, money & measure	maths.js	multi
Percentages	Number, money & measure	maths.js	multi
Algebra	Expressions & equations	maths.js	multi
Area, perimeter & volume	Number, money & measure	maths.js	multi
Angles	Shape, position & movement	maths.js	multi
Order of operations (BODMAS)	Number processes	topics-varied.js	6
Fractions	Fractions, decimals & %	topics-varied.js	7
Decimals	Fractions, decimals & %	topics-varied.js	6
Ratio & proportion	Number, money & measure	topics-varied.js	6
Negative numbers	Number processes	topics-varied.js	6
Averages & data	Information handling	topics-varied.js	7
Time & speed	Number, money & measure	topics-varied.js	5
Multi-step problems	Problem solving (all organisers)	topics-varied.js	7
Multi-step challenges	Problem solving (all organisers)	challenges.js	8

"multi" means several styles chosen inside the generator. Diagrams come from **visual.js** (spoiler-free SVG builders), and names and pronouns from **names.js**.

How every question is checked

- **Answers verified independently:** BODMAS by evaluating the expression, angle sums, percentages, areas and algebra solutions.
- **Sweeps on every build:** every topic × style × tier × dozens of seeds must have exactly one correct option, no two options with the same value, no float artefacts, no "NaN"/"undefined", and pronouns that match names.
- **Spoiler checks:** no diagram labels the answer, and no hint contains the spelling word or a 5+ letter chunk of it.
- **Hand review:** 983 literacy items and 252 maths bank items were read by a reviewer; every issue found is fixed and locked in with a regression test (1,079 tests in total).

Maths • item banks 252 HAND-WRITTEN ITEMS

TOPIC	CFE ORGANISER	SOURCE	ITEMS
Fractions, decimals & %	Fractions, decimals & %	items/maths-expanded(-extra).js	23
Money & coins · Unit conversions	Money · Measurement	items/maths-expanded(-extra).js	15 · 18
Area & perimeter · Volume · Angles	Measurement · Angle, symmetry	items/maths-expanded(-extra).js	13 · 13 · 13
Coordinates · 3-D shapes	Shape, position & movement	items/maths-expanded(-extra).js	15 · 13
Mean, median & range	Information handling	items/maths-expanded(-extra).js	13
Multiples & factors · Ratio · Rounding	Number, money & measure	items/maths-expanded(-extra).js	13 · 12 · 13
Sequences · Using formulae	Patterns · Expressions & equations	items/maths-advanced.js	20 · 19
Probability · Reading graphs & charts	Chance & uncertainty · Data	items/maths-advanced.js	15 · 24

Spelling 298 WORDS · 24 SPOT-THE-SPELLING ITEMS

TOPIC	SOURCE	WORDS	MC
Homophones & confusables	spelling.js + items/spelling-extra.js	95	5
-ible and -able · -ant/-ent, -ance/-ence	same	37 · 37	4 · 3
Silent letters & doubles · Greek & Latin roots	same	47 · 37	4 · 3
Commonly misspelled	same	45	5
This week's words (family's own list)	typed in by a grown-up, read aloud	—	—

Writing & Grammar 501 ITEMS

SOURCE FILE	TOPICS	ITEMS
grammar.js (base)	all core topics	32
clauses-voice.js	clauses 60 · voice 40	100
punctuation-cohesion.js	punctuation 80 · cohesion 50	130
tense-register-wordclass.js	tense 45 · register 40 · word class 45	130
grammar-sats.js	verbs, contractions, Standard English, determiners, adverbials, modals, pronouns, reported speech, word families	100
passages.js	3 passages · comprehension	9

Vocabulary 172 ITEMS

TOPIC	SOURCE	ITEMS
Synonyms · Antonyms	vocab-items.js	30 · 28
Words in context	vocab-items.js	28
Word classes · Morphology	vocab-items.js	26 · 26
Formal & informal register	vocab-items.js	25
Reading comprehension	passages.js	9

Passages: The Storm at Portree · The Falkirk Wheel · A Ceilidh in Tobermory · Nadia's Market Stall · Puffins on the Isle of May · Midges at Loch Lomond